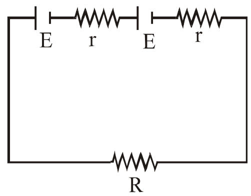
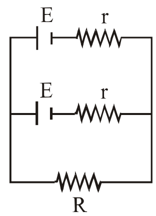
Solutions:-

1. 



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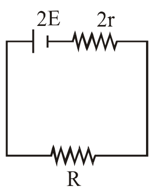
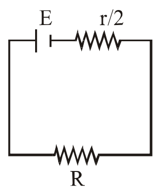


R = 2r



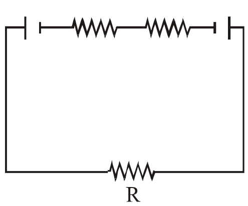
R = r/2

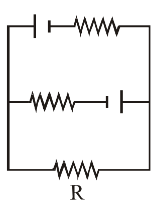
$i = \frac{2E}{4r} = \frac{E}{2r}$ $\qquad$ $i = \left(\frac{E}{r}\right)$

$P_d = i^2(2r)$ $\qquad$ $P_d = i^2\left(\frac{r}{2}\right)$

$= \left(\frac{E^2}{2r}\right)$ $\qquad$ $P_d = \left(\frac{1E^2}{2r}\right)$

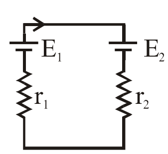
Opposite polarity





$P_d = 0$ $\qquad$ $P_d = 0$

2. $I = \frac{E_1 - E_2}{r_1 + r_2}$



Potential difference across cells will be same and cannot be zero.

3. Number of free e^- per unit volume of metal is very large as compare to number of free ions per unit volume of electrolyte

$\sigma = \left(\frac{ne^2\tau}{m}\right)$

$\sigma \propto n$

4. Materials like Nichrome, Manganin and constantan exhibit a very weak dependence of resistivity with temperature and are thus used in wire bound standard resistors.

5. Rating of the bulb is done at steady state of the filament when filament is at room temperature its resistance is less than its resistance when the bulb is at steady state (fully illuminated)

6. $\rho = \left(\frac{m}{ne^2\tau}\right); \sigma = \frac{ne^2\tau}{m}$

$\rho \propto \frac{1}{n} \qquad \sigma \propto n$

In electrical conduction only free e^- take part so value of n is small & conductivity is small but in thermal conduction both lattice and free e^- contributes

∴ value of n will be greater

∴ σ will be greater

7. Resistance decreases due to decrease in temperature, so current through wire increases.

8. When 20 bulbs are in series

$i = \left(\frac{V}{20R}\right)$

$(P_d)_1 = (i^2R)20 \Rightarrow \frac{V^2}{20R}$

When 19 bulbs are in series

$i = \left(\frac{V}{19R}\right)$

$(P_d)_2 = (i^2R)19 = \left(\frac{V^2}{19R}\right)$

$(P_d)_2 > (P_d)_1 \Rightarrow$ Brightness ↑

9. Since potential difference between two points in a circuit does not depend on path therefore one can conclude that electrostatic field is conservative field

10. Copper has positive temperature coefficient where as germanium has negative temperature coefficient

11.

$I = neA\,V_d$	$\rho = \frac{m}{ne^2\tau}$
$\frac{V}{R} = neA\,V_d$	on ↑ T ⇒ τ ↓
$\frac{V}{\rho\ell}A = ne\,AV_d$	⇒ ρ ↑
	⇒ V_d ↓

$V_d \propto \frac{1}{\rho}$

12. $I = neAV_d$

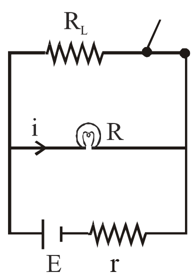
$$\frac{V}{R} = neA\,V_d$$

$$R = \left(\frac{neAV_d}{V}\right)$$

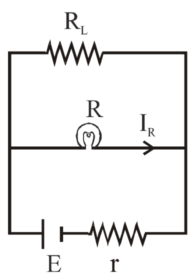
Since no. of e^-/vol, area of cross section ,applied potential difference & drift velocity remain same

$\therefore$ Resistance will not change.

13.



$$i = \left(\frac{E}{R+r}\right)$$



$$I_R = \frac{E}{r + \frac{RR_L}{R+R_L}} \times \left(\frac{R_L}{R+R_L}\right)$$

$$I_R = \frac{ER_L}{rR + rR_L + RR_L}$$

$$I_R = \frac{E}{r + R + \frac{rR}{R_L}}$$

$i_R \downarrow \Rightarrow$ Brightness $\downarrow$

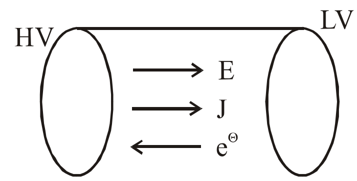
14. In non-conductors, Insulator and wood electrons are bound which will not accelerate on application of electric field. In conductors free electrons are present .Which when electric field is applied, they accelerate and electric current is generated

15. Electrolyte has both positive and negative charge carriers.

16. There will be steady electric field in body of conductor. This will result in a continuous current rather than a current for short period of time. Mechanism which maintain a steady electron field are cells and Batteries

17. I/A = J (Current Density)

18.



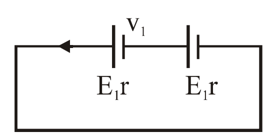
$$\vec{J} = \sigma\vec{E}$$

19. As electron number density in solid conductor is enormous $\rightarrow 10^{29}\ m^{-3}$

20. If charges moved without collisions through the conductor, their kinetic energy would also change so that the total energy is unchanged conservation of total energy would then imply that,

$\Delta K = -\Delta U_{pot}$ i.e., $\Delta K = I\,V\Delta t > 0$

21. If an identical battery is connected in supportive Net emf = 2ε



$I = \frac{2\varepsilon}{2r} = \frac{\varepsilon}{r}$, So potential difference across any battery $v_1 = \varepsilon - Ir = \varepsilon - \frac{\varepsilon}{r} \times r$ $\boxed{v_1 = \varepsilon - \varepsilon = 0}$

22. Loop Rule- The algebric sum of changes in potential around any closed loop involving resistors and cells in the loop is zero. this rule is also obvious, since, electric potential is dependent on the location of the point. Thus starting with any point, if we come back to the same point, the total change must be zero. In a closed loop, we do come back to the starting point and hence the rule. $\Sigma Ir = \Sigma V$

23. In meter bridge balanced wheat stone bridge is used to determine unknown resistance.

24. The potentiometer has the advantage that it draws no current from the voltage source being measured.As such it is unaffected by the internal resistance of the source.

25. The working of potentiometer is based on the fact that the fall of potential across any portion of the wire is directly proportional to the length of that portion provided the wire is of uniform area of cross-section and a constant current is flowing through it.To shift the balance point on higher length, the potential gradient of the wire is to be decreased. The same can be obtained by decreasing.The current of the main circuit which is possible by increasing the resistance in series of potentiometer wire.

26. When we measure the emf of a cell by the potentiometer then no current flows in the circuit is zero- deflaction condition i.e., cell is in open circuit.Thus, in this condition the actual value of emf of a cell is found. In this way, potentiometer is equivalent to an ideal voltmeter of infinite resistance.

27. In meter bridge for measurement of resistance the known and the unknown resistance are inter changed. The error so removed is end correction.

28. The relationship between current and drift speed is given by $I = neA\,V_d$ Here, I is the current and V_d is the drift velocity. so, $I \propto V_d$

29. Kirchhoff's junction rule is also known as kirchhoff's current law which states that the algebric sum of the currents flowing towards any point in an electric network is zero i.e., charges are conserved in an electric network so, kirchhoff's junction rule is the reflection of conservation of charge.

30. Current always flow from HV$\rightarrow$ LV.

31. The electrical resistance of a conductor is depends upon all factors size, temperature and geometry of conductor.

32. $V_d = \dfrac{I}{nAe} = \dfrac{J}{ne} = \left(\dfrac{\sigma}{ne}\right)E$.

33. V = IR is equation of straight line.

34. The figure is showing I-V characteristics of non-ohmic or non linear conductors.

35. Cu 1.7×10^{-8}
Nichrome 100×10^{-8}
Germanium 0.46
Silicon 2300

36. Temperature coefficient of metals is positive and that of semiconductors is negative. Therefore with increases in temperature, conductivity of metal decreases and that of semiconductor increases.

37. Initially temperature of heater is low. but later its temperature increases and resistance increases $\therefore$ Initial current is slightly higher than steady state value.

38. For series combination current is same.

39. Null deflection principle is used in potentiometer.

40. No change happens due to exchange.

41. All statements are correct.

42. Due to parallel combination resistance decreases and current increases.

43. Due to increase in resistance(R) the current decreases, hence potential gradient decreases, Therefore balancing length increases,so J will shift towards B.